

SECTION 01 11 00

STATEMENT OF OBJECTIVES

PART 1 GENERAL

1.01 PROJECT BACKGROUND

The Bureau of Reclamation, Upper Colorado Region, has a requirement for a hybrid Fixed Price and Progressive Design-Build for the Pojoaque Basin Regional Water System (PBRWS) Phase 2. The Pojoaque Basin Regional Water System (PBRWS) was authorized by the Aamodt Litigation Settlement Act (P.L. 111-291, Title VI) (Act). The Act established several objectives with deadlines and is being procured in phases.

- The PBRWS is to have an ultimate capacity of delivering 4,000 acre-feet per year with a peak daily estimated demand of 6.23 million gallons per day (MGD).
- Substantial completion of the project will be the building of portions of the system that can provide an initial anticipated demand of 2,500 acre-feet per year with components capable of accommodating the ultimate 4,000 acre- feet per year.
- PBRWS will have the ability to transfer up to 1,000 acre-feet per year of unused water from the lower sections of the system to a connection tee at the intersection of Bishop's Lodge Road and Tesuque Village Road.
- The PBRWS will provide drinking water that meets or exceeds safe drinking water act standards.
- Water is to be supplied to users within San Ildefonso, Pojoaque, Nambé, and Tesuque Pueblos; and a portion of Santa Fe County (a.k.a. Bishops Lodge extension).
- PBRWS Phase I is currently being constructed under contract R17PC00023.
- This contract action is undertaken within the authority of Reclamation which was authorized by Congress for the design and construction of The Pojoaque Regional Water System (PBRWS) within the state of New Mexico

1.02 DESCRIPTION OF WORK

1. The Project is to be designed and constructed using a hybrid Fixed Price and Progressive design-build delivery method. Some principal components of work are well-defined and shall be bid up front following a Fixed Price design-build delivery method, while other components of work are not well-defined and shall follow the Progressive design-build delivery method.
2. This contract differs from a typical design-build contract in that designs have already been completed for this project to the 60-100% level. These designs may require some substantial revisions, returning some elements to the 0-30% level.
3. Design Phase: Utilize existing design drawings and WaterGEMS model information at varying percentages of completion (e.g. conceptual, 30%, or 60%), collect necessary additional design data, incorporate value engineering and innovative technology to prepare design to 90% complete. For Principal Components of Work following a progressive bid schedule, the contract will apply an iterative design and cost-estimation process. The contractor will produce

a construction cost estimate at 30% and 60% design levels with iterative feedback from Reclamation to guide the project to an acceptable scope and price. At 90% the contractor shall develop a fixed price proposal for construction activities.

- a. The best practice for this progressive iteration of scope and cost development is to not progress to the next phase of design until there is agreement on the current scope and budget. This discipline requires that project features be adjusted, owner preferences prioritized, and construction costs verified.
4. Construction Phase: Complete 100% design and construction tasks, including construction inspection services, cultural monitoring services, system disinfection, and as-built drawings (if firm-fixed price is approved by the Owner at the end of the Design Phase). The Construction Phase also includes incorporating specified existing drinking water infrastructure into the Project.
5. Post-Construction Phase: Complete pump station testing, demonstration testing, operator training and provide technical support during the initial 6 months of Project operations. Project operations will be performed by the County of Santa Fe on behalf of the Pojoaque Basin Regional Water Authority.
6. Even when the Contractor elects to use Government-provided 100% design documents, the Contractor shall remain fully responsible for the final design and shall assume all duties and liabilities of the Engineer of Record.

1.03 OBJECTIVES

- A. The Owner's objectives for delivery of the Project described below are as follows:
 1. **Quality**: Provide facilities and equipment that fully integrate with existing features of the Pojoaque Basin Regional Water System in full compliance with federal and state regulations and contractual standards for water quality. The project needs to be designed and constructed to operate through a wide range of production quantities, while delivering water that meets federal clean drinking water standards for all customers, varying from 170,000 gallons per day on average for the first three to five years of operation to an estimated 6.23 million gallons per day to meet ultimate peak daily demand. The Phase 1 (CDM Smith) and Phase 2 projects of the PBRWS shall be compatible so the system can be operated together as a single regional water system. The system will incorporate fire protection. Design and construction shall comply with all environmental and cultural resources commitments required by federal, state, local and tribal permits. Cooperation and communication are needed to minimize road closures and utility outages during construction.
 2. **Schedule**:
 - a. Achieve stakeholder facilitation and review of 90% design of the prioritized construction line items in Price Schedule C to begin construction within 180 days after contract award. Completion of design, construction, commissioning, and performance testing of the project

- components within the remaining 1,120 days after the start of construction.
 - b. The construction schedule shall be approved by the Contracting Officer (CO) as stated in FAR 52.236-15.
 - 1) The CO will prioritize completion of salient features of work as necessary, if it is in the best interest of the government.
 - 3. Risk: Achieve an optimal balance of risk allocation between the Owner and the Design-Builder.
 - 4. Safety: Implement an effective safety program incorporating best industry practices that meet or exceed all relevant Reclamation Safety and Health Standards (RSHS).
- B. By selecting a hybrid Fixed Price and Progressive design-build delivery method for the Project, Reclamation and stakeholders are committed to working in close collaboration with the Design-Builder during the Design Phase to develop the Project's design to achieve the Project objectives and to obtain a mutually agreeable lump-sum price for delivery of the Project that meets this Project's available budget. As set forth in this Statement of Objectives (SOO), the Owner has certain technical requirements and standards that will apply to the Project's design.

1.04 LOCATION

- A. The project is located approximately 15 miles north of the city of Santa Fe, New Mexico. The project footprint extends onto the lands of Nambe Pueblo, Pojoaque Pueblo, Tesuque Pueblo and throughout Santa Fe County.

1.05 DEFINITIONS

- A. When specifications use a word or term defined in Federal Acquisition Regulations (FAR) <https://www.acquisition.gov/browse/index/far>, definition of the word or term shall be in accordance with FAR sections in effect at the time solicitation was issued.
- B. "Act" means Title VI of Public Law 111-291, <https://www.congress.gov/111/plaws/publ291/PLAW-111publ291.pdf>, enacted on December 8, 2010, which may be cited as the "Aamodt Litigation Settlement Act".
- C. "Cost Sharing and System Integration Agreement" means the agreement dated August 27, 2009, executed by the United States, the State, the Settlement Pueblos, the County, and the City on March 14, 2013, that describes the PBRWS and allocates the costs of the PBRWS.
- D. "County Distribution System" means the portion of the PBRWS that serves water customers on non-Pueblo land in the Pojoaque Basin.
- E. "Cultural Resources" include historical properties, archaeological resources, and traditional cultural properties.

- F. “Engineering Report” means the report entitled, “Pojoaque Regional Water System Amended Engineering Report” dated December 2019, prepared by United States Department of the Interior, Bureau of Reclamation, Technical Services Center pursuant to 2019 section 611(g) Agreement of the Settlement Act. This document amends, supersedes, and replaces the Engineering Report referenced in the Settlement Act.
- G. **SUPERSEDED** “Engineering Report” means the report entitled, “Pojoaque Regional Water System Engineering Report” dated September 2008, prepared by Hurlbut, Kersich, and McCullough (HKM) Engineering for the Northern Pueblos Tributary Water Rights Association, and any amendments thereto, including any modifications which may be required by section 611(d)(2) of the Act.
- H. “Final Owner” means the entity who will receive title to a project feature or pipeline segment upon substantial completion of the PBRWS and in accordance with the Act.
- I. “Operating Agreement” means the agreement between the Pueblos and Santa Fe County executed under section 612(a) of the Act.
- J. “Owner” means the Bureau of Reclamation (Reclamation), who is responsible for planning, designing and constructing the Project on behalf of the Pueblo de San Ildefonso; Pueblos of Pojoaque, Nambé and Tesuque; and Santa Fe County.
- K. “PBRWS” means the Pojoaque Basin Regional Water System described in section 611(a) of the Act.
- L. “Pueblo Water Facilities” means;
1. A portion of the PBRWS that serves only water customers on Pueblo land; and
 2. Portions of a Pueblo water system in existence on the date of enactment of this Act that serve water customers on non-Pueblo land, also in existence on the date of enactment of this Act, or their successors, that are--
 - a. Depicted in the 100% project design, as modified by the drawings reflecting the completed Regional Water System; and
 - b. Described in the Operating Agreement.
- M. “Stakeholders” means the Pueblo de San Ildefonso, Pueblo of Pojoaque, Pueblo of Nambé, Pueblo of Tesuque, Santa Fe County, Pojoaque Regional Water Authority, New Mexico Office of the State Engineer, Bureau of Indian Affairs, Indian Health Service, and New Mexico Department of Transportation, Environmental Protection Agency.

1.06 PHASE 1 (CONTRACT R17PC00023)

- A. The intake and water treatment plant of Phase 1 (CONTRACT R17PC00023) of PBRWS is capable of diverting, treating, and delivering from 566.4 acre-feet up to 4,000 acre-feet of water annually (afy) to end users, including households and businesses, in the Pueblo de San Ildefonso and Pojoaque, Nambé, and Tesuque Pueblos (the Settlement Act Pueblos) and portions of Santa Fe County north of the City of Santa Fe, New Mexico.

The system will collect water from the Rio Grande, transport the untreated water to a treatment facility, transmit the treated water to water storage/distribution tanks, and then distribute the treated water to end users. The Phase 1 contractor is responsible for all features hydraulically dependent up to, and including, the pipelines to the Nambé Pump Station and up to, and including, the forebay tank for South Pojoaque Pump Station.

1.07 PRINCIPAL COMPONENTS OF WORK

This Phase 2 contract includes design and construction of but is not limited to:

A. System Modeling and Report –

1. Existing designs: Reclamation's TSC has completed a WaterGEMS hydraulic model that incorporates the Phase 1 and the Phase 2 Design. Reclamation's TSC has also completed Phase 2 designs and specifications to the 60-100% level. Phase 1's Design Builder has completed 100% Phase 3 distribution line designs for San Ildefonso and parts of Pojoaque. The contractor should use this model, the drawings, and specifications contained in the appendices as a guideline to develop the most economic and robust water system that ultimately satisfies the AAMODT Litigation.
2. New requirements: There is not anticipated to be enough funding to complete the entire Phase 2 as described in the above available existing designs and reports. The stakeholders have eliminated some features that they are no longer interested in, and the remaining features are listed as principal components of work B-J below.
 - a. The model shall determine if the remaining features are hydraulically adequate and re-design the system to compensate for fatal flaws, if any, resulting from the missing features.
 - b. The final design shall anticipate that the Regional Water Authority will make expansions after Phase 2 and shall include modeling to ensure future expansions are hydraulically compatible with the current designs.
 - c. The deliverable shall be a design report describing the results of the modeling.
 - d. Cuyumungue Expansion– In addition to the general Phase 2 modeling efforts described above, there will specifically be a feasibility investigation of providing water delivery to the Cuyumungue neighborhood in the southeast corner of Pojoaque Pueblo. The Phase 2 tank intended to serve this neighborhood was eliminated by stakeholders. This effort would investigate the feasibility of alternative methods for serving this neighborhood from the existing Buffalo Thunder Tanks or Turtle Tank. This effort would include leading meetings with stakeholders, data collection and modeling, and presenting 30% design alternatives and cost estimates for stakeholder decisions.

B. County Interconnect – approximately 15 miles of pipeline, 3 pump stations, and 2 forebay tanks. This component is a transmission pipeline to transmit potable water from

the Phase 1 South Forebay Tank on Tribal Works Rd in Pojoaque to a point of future interconnection with Santa Fe County's water distribution system at the intersection of Tesuque Village Rd and Bishop's Lodge Rd. A distribution line shall be buried in the same trench. The transmission system shall be capable of reverse flow, bringing water from the County's system into the Pojoaque Basin Regional Water System.

1. Pipeline:

- a. Alignment: the pipeline alignment shall follow the existing alignment design, as shown in the 100% P2S2 Pipelines design, with two exceptions:
 - 1) The pipeline shall be realigned to follow the US-285 frontage road starting at Camel Rock studios until the intersection of US-285 and Tesuque Village Rd.
 - a) There is no utility or Geotech data for this stretch.
 - 2) Other re-alignments are allowed for value engineering purposes with the permission of the respective Pueblo. Private land easements are time-extensive to obtain and thus it is recommended to value engineer re-alignments to avoid private parcels.
- b. Design Data: Geotech and utility information for the existing pipeline alignment is available.
- c. Transmission and distribution details:
 - 1) The distribution pipeline shall include appropriately spaced fire hydrants, except in the segment from the northern border of Tesuque Pueblo to the intersection of US-285 and Tesuque Village Rd.
 - 2) Provide distribution tees at all locations where future system expansion would reasonably be anticipated for a water utility.
 - 3) Seven transmission tees shall be provided at locations including Buffalo Thunder tank/Nambe Falls Travel Center, Camel Rock Casino tank, Tesuque IGC tank, Tesuque Oweenge Day School tank, Tesuque Mobile Home Park tank, Tesuque Casino tank, and at the intersection of Tesuque Village Rd and NM-592.
- d. Timeline: The County Interconnect is highest priority to complete as soon as possible, and the contractor shall prioritize the design and construction over all other principal components.

2. South Pojoaque Pump Station:

- a. At this location, the forebay tank and associated site work has been completed by the Phase 1 contractor. The remaining work is to build the Pump Station and associated site work, as shown in the 100% P2S2 Site Facilities design.

3. Camel Rock Pump Station:

- a. This site includes both a forebay tank and pump station. The forebay tank and pump station have been designed to 100% as shown in the P2S2 Site Facilities drawings. The Pueblo of Tesuque has indicated that it would prefer the pump station to be relocated to the western side of US-285. The Pueblo has approved a site north of the old RV Park on the west side of US-285, approximately 1,200 feet northwest and 40 feet lower in elevation. Geotech information is not available for this site.
- 4. Tesuque Pump Station:
 - a. This site includes both a forebay tank and pump station. The forebay tank and pump station have been designed to 100% as shown in the P2S2 Site Facilities drawings. The Pueblo of Tesuque has indicated that it requires the pump station to be relocated to the southeast area of the intersection between US-285 and Tesuque Village Rd. This new site is approximately 2,000 feet northwest and 30 feet lower in elevation. Geotech information is not available for this site.
 - b. During the design phase, prepare cost estimates to compare burying the forebay tank, partially burying the forebay tank, and not burying the forebay tank. For bidding purposes, assume the tank will not be buried.
- C. Nambe Pueblo Connection – Approximately 3.5 miles of pipeline, 1 pump station, 1 forebay tank, and one connection to an existing tank. This component is a transmission pipeline to provide potable water to the Pueblo of Nambe from the point where Phase 1 leaves off on NM-503. A distribution line shall be buried in the same trench.
 - 1. Nambe Pump Station: This site on the intersection of Don Bernardo Rd and NM-503 includes both a forebay tank and pump station. The forebay tank and pump station have been partially designed as shown in the P2S1 drawings.
 - 2. Nambe Pueblo Connection Pipeline:
 - a. The pipeline will include both a transmission and distribution line in the same trench. The starting point of the pipeline is the Nambe Pump Station, and the ending connection point to an existing tank is TBD. The contractor shall provide a 30% design and cost estimate for two alternative destinations described below. Once the Pueblo selects the alternative, the contractor shall proceed with design to 100%.
 - 1) Alternative Destination #1 - Nambe Buffalo Tank: The 100% P2S1 designs included a pipeline reaching the Nambe Buffalo Tank site. The contractor shall value engineer the pipeline alignment to reach the Nambe Buffalo Tank but avoiding the culturally sensitive Pueblo Plaza.
 - 2) Alternative Destination #2 - New Nambe Tank: The contractor shall value engineer the pipeline alignment to reach a new conceptual Nambe tank, likely in the northern portion of the Pueblo near NM-503.

- a) Due to the large amount of uncertainty on this conceptual tank, the 30% design and cost estimate for this alternative will not be due until 3 months after Reclamation provides the site information for the new tank.
 - b. This component also includes approximately 540 feet of pipeline to get from the end point of the Phase 1 pipeline on NM-503 to the Nambe Pump Station. This pipeline segment is shown on the 100% P2S1 drawings.
 - c. Transmission and distribution details:
 - 1) The distribution pipeline shall include appropriately spaced fire hydrants.
 - 2) Provide distribution tees at all locations where future system expansion would reasonably be anticipated for a water utility.
 - 3) Provide a transmission tee for future Pueblo expansion to the alternative pipeline destination that was not selected.
 - 4) Utility and Geotech data are available along the current alignment.
- D. Buffalo Thunder Tank Connection – Approximately 3,200 feet of pipeline to supply potable water to the Buffalo Thunder Tanks on the south side of the Buffalo Thunder Resort and Casino from the nearest point on the “County Interconnect” pipeline.
 - 1. This Buffalo Thunder Tank Connection pipeline and connecting features are shown in the 100% P2S2 Pipelines drawings. The pipeline may be value-engineered to shorten the pipeline distance while still providing a connection point to the Nambe Development Area existing pipeline network (south and east of Nambe Falls Travel Center).
- E. County Distribution Phase 3 – Approximately 15-20 miles of distribution pipelines designed to ceiling of \$20.4 million (including contingency). Locations of the distribution pipeline to be determined based cost effectively maximizing connections to customers who have elected to connect to the system. Location data for customers who have signed up to connect will be provided during design.
- F. Yellowbird East Pipeline – Approximately 10,538 feet of distribution pipeline in the Phase 1 geographic area completing a loop between two existing pipe segments.
 - 1. Yellowbird Road is a proposed County road currently in the final stages of design. The Phase 1 contractor completed 100% design alignment for this road but the vertical alignment is not completed. Once the County finalizes the roadway designs and puts the construction of the road out to bid, the Phase 2 contractor shall finalize the pipeline designs. Once the County roadway contractor constructs finished grade of the roadway, then the Phase 2 contractor shall install the pipeline. After that, the County roadway contractor shall complete paving activities.
- G. RKM Pipeline – Approximately 14,000 feet of transmission pipeline starting from the point of County Interconnect and continuing to a new distribution tank to provide potable

water to the Tesuque Village neighborhood. A distribution pipeline will be buried in the same trench.

1. The pipeline alignment shall generally follow the existing alignment design, as shown in the 60% P2S3 drawings.
2. Utility and Geotech data will need to be collected along the alignment.

H. RKM Tank – A new distribution tank to provide potable water to the Tesuque Village neighborhood.

1. Utility and Geotech data will need to be collected at the RKM tank site (currently owned by project partner Santa Fe County).
2. The contractor shall work with Reclamation to determine the size of the tank considering available funding. The originally proposed tank size is shown in the 60% P2S3 drawings and this size shall be used for initial bidding purposes.

I. New Nambe Tank – A new tank at Nambe next to the existing Nambe Buffalo Tank to provide additional distribution storage for Nambe Pueblo's existing distribution system as well as for County residents within the boundaries of Nambe Pueblo.

1. This component includes connecting into the transmission pipeline at the Nambe Buffalo Tank.
2. The contractor shall work with Reclamation to determine the size of the tank considering available funding. The originally proposed tank size is shown in the 100% P2S1 drawings.
3. Geotech data is available in the 100% P2S1 package.

J. Tesuque Casino Tank Connection – Approximately 3,000 feet of transmission pipeline to provide potable water to the Tesuque Casino.

1. Transmission pipeline shall connect from the Tesuque Village Road transmission pipeline (built under this contract) to the connection point at the southeast corner of the Tesuque Casino parking lot. The currently planned alignment is shown in the 100% P2S2 Pipelines drawings.
2. Pipeline alignment may be value engineered. The constraints include not disrupting Tesuque Casino business during construction in a manner satisfactory to the Casino management under the supervision of the Pueblo.

K. SCADA

1. Integrating Phase 2 into the existing Phase 1 SCADA system ensuring adequate communication and appropriate repeater sites between the RKM Tank site and the existing Turtle Tank site.
2. Site Facilities include but are not limited to:
 - a. Nambe Pumping Station and Forebay Tank
 - b. Nambe Tanks #25 and #27

- c. Pojoaque South Pumping station and Forebay Tank
 - d. Buffalo Thunder Tank #22
 - e. Camel Rock Pumping Station and Forebay Tank
 - f. Tesuque Pumping Station and Forebay Tank
 - g. RKM Tank
- L. This contract includes services such as but not limited to:
- 1. Facilitator
 - a. Contractor shall facilitate forums to present proposed design and cost information to stakeholders and facilitate discussions to ensure stakeholder consensus.
 - b. Location of forums shall be determined by the contractor and approved by the CO/COR.
 - 2. Constructability Review
 - a. The design phase must include a field constructability review.
 - 3. Cultural Resource Monitoring Services
 - a. Contractor shall employ an independent Cultural Resource Monitoring Services provider to provide monitoring during subsurface exploration during the design and construction phases of the project.
 - 1) This independent Cultural Resource Monitoring (CRM) Services provider shall also be responsible for working with the four Pueblos' Tribal Historic Preservation Offices and the State Historic Preservation Office to develop or update Archeological Treatment Plans and NAGPRA Plans of Action for the construction activities. Most of the project area has had archeological surveys completed but if a new alignment is proposed, the CRM provider may be required to perform new archeological surveys. This CRM provider will advise the designers on culturally sensitive areas to avoid.
 - 4. Construction Inspection Services
 - a. Contractor shall employ an independent Construction Inspection Services sub-contractor to provide construction inspection during the construction phase of the project.
 - b. The sub-contractor shall oversee daily onsite construction activities and prepare daily inspection reports for compliance with drawings, specifications, and applicable safety standards. At a minimum, inspection staff include two (2) civil inspectors, 1 mechanical technician/inspector and 1 electrical technician/inspector. An inspector shall be on site for a minimum of 50% of daily activities for each crew or site and 100% during critical activities. Critical activities to include installation of all electrical

- equipment, conduit and conductors, installation of all mechanical equipment (does not include pipelines), excavation and earthwork for tanks and building foundations, concrete placements, road crossings (open or jack and bore), any work in the NMDOT Right of Way, tank erections, testing of all equipment to include hydrostatic testing, electrical equipment checks, mechanical checks, preliminary testing of electrical and mechanical equipment and commissioning.
- c. Mechanical technicians/inspectors must have a minimum of 5 years' mechanical experience in construction and installation of pumping plants or water treatment plants and tank installations, installation of miscellaneous mechanical equipment and testing of mechanical equipment.
 - d. Electrical technician/inspector must have a minimum of 5 years' electrical experience in construction of high voltage equipment in pumping plants or water treatment plants, conduit and conductor installation and testing of conductors and electrical equipment.
 - e. Civil inspectors must have a minimum of 5 years' experience in the inspection of pipelines, earthwork foundations, concrete placements and industrial building erection.
 - f. Contractor to submit to the Bureau of Reclamation the resumes of the inspectors/technicians for approval. Inspector's position is a neutral position that documents the construction. The Contractor shall not staff the project with any person who is current or former employee of Sante Fe County, City of Santa Fe, NMDOT, or any of the stakeholders including the Pueblos to the PBRWS project. Employment selections shall avoid conflicts of interest or the impression of one.
 - g. Inspectors/technicians to be proficient in typed digital reports. At a minimum, the daily report will include: contract name, contract number, shift, date, weather, contractor and contractor's report, subcontractor, inspector's signature, safety, discussions, environmental, description of activities, labor hours (class, hours worked and description of work for every employee), equipment hours (equipment description, quantity, hours operated, hours down, hours idle and description of work) and two (2) photos per site. Attachment A is an example of an approved inspection daily report. Daily reports to be submitted to the Bureau of Reclamation's construction manager and contracting officer's representative by 0900 the next day. A weekly construction report with a summary of the week's construction activities will be submitted to the Bureau of Reclamation's construction manager and contracting officer's representative by 0900 the Wednesday of the following week.
5. Utility Coordinator
- a. Contractor shall be responsible for facilitating coordination and acquiring all necessary utility permits.

- b. The contractor shall be responsible for completing and submitting applications. Cost incurred for design and construction of utility extension will be paid by the contractor. Reimbursement for the cost incurred shall be included in the appropriate line item of the proposal.
 - 6. System Commissioning
 - a. Contractor shall be responsible for commissioning PBRWS Phase 2 in coordination with the operational Phase 1.
 - b. Contractor shall provide Technical Support during commissioning and transitional operation to ensure system functionality.
 - 7. As-builts shall be provided for inspection during construction and final as-builts shall be provided upon project completion.
 - 8. Training
 - a. The contractor shall provide training for Phase 2 features of PBRWS operational staff and a detailed Standard Operating Procedure manual to ensure seamless operation in perpetuity.
 - 9. Operational Tools Update
 - a. The contractor shall update the Phase 1 O&M Plan to include Phase 2 considerations.
 - b. The contractor shall update the Phase 1 CMMS to include the appropriate preventative maintenance actions for Phase 2
 - c. The contractor shall update the existing GIS and SCADA with Phase 2 information to facilitate ease of operations and maintenance
- M. Phase 1 and Phase 2 portions of the PBRWS are intended to be operated together, as a single regional water system. The Contractor shall coordinate with Reclamation and Phase 1 - CDM Smith's design-build team to meet this objective.
- N. Incorporation of Existing Infrastructure
 - 1. In addition to the new facilities being designed and constructed under this contract, existing drinking water infrastructure, described in Section 01 56 60 of this document, shall be incorporated into the new regional water system, as a cost-efficient means of providing treated water and fire protection to the service area included in this contract.
 - 2. Where existing infrastructure has been determined to be insufficient to accommodate fire protection, the Contracting Officer will make a determination if it is in best interest of the government to modify the contract to replace existing infrastructure in Phase 3 of the project.
- O. The Design-Build contractor shall seal drawings and specifications with a current New Mexico Professional Engineer License.
- P. System components

1. Transmission
 - a. The water transmission system would consist of buried pipelines, pump stations, distribution tanks, and appurtenant items. The contractor would be responsible for all features hydraulically dependent beyond, and including, the Nambé Forebay Tank Pump Station, and including, the Pojoaque South Pump Station (Pojoaque South Forebay Tank is included in Phase 1). The design of these pre-fabricated pump stations shall be compatible with the Phase 1 design-build pipeline design.
 - b. The transmission system includes the pipelines from the forebay tanks to the distribution tanks. Transmission water would be lifted by in-line pump stations to the next pump stations and distribution tanks.
2. Tanks
 - a. Pump stations will have a forebay tank directly upstream.
 - b. Storage/Distribution tanks would be at the end of the transmission lines to supply water to the end users.
3. Distribution
 - a. Water will be delivered from the storage/distribution tanks to the end users via distribution pipelines. The Phase 1 distribution systems are predominantly looped systems within pressure zones, and each pressure zone may be isolated from the other zones by pressure reducing valves. The pressure reducing valves would be 'closed' during normal operations but may be 'opened' during emergency situations when larger volumes of water are required allowing for water sharing across pressure zones.
 - b. The distribution tanks and distribution pipelines should be sized for the fire protection requirement of one hour at 1000 gallons per minute (gpm) flow. See Section 2.4 in the Amended Engineering Report which defines the sizing of tanks.
4. Pump Stations
 - a. The Phase 2 design will include up to 4 pump stations: one in the Phase 2 Stage 1 Nambe Pueblo system to transmit water from the Turtle Tank transmission system to the distribution tank locations in east Nambe and three in the Phase 2 Stage 2 Tesuque Pueblo/Santa Fe County Interconnect to transmit water from Pojoaque South pump station to RKM Tank. Air chambers will be required immediately downstream of each pump station. An electrical power transformer would also be located adjacent to the pump station facility. Pump station buildings will be designed to allow for operations, maintenance, and replacement (OM&R) activities. Locations for the pump stations are identified in the existing drawings. Relocation of the pump stations shall be considered in the design phase of this contract.
5. Pipe Trenches

- a. Distribution and transmission mains shall be placed in joint trench wherever feasible and economical. This practice will aid in the reduction of time that roads need to be closed and increase the productivity of pipe laying. Consideration is needed during design to facility backfilling based on pipe sizes and pipe spacing.
6. Pipe Naming Convention
 - a. Pipeline naming conventions include the 3-letter appendix to indicate if the pipeline wholly serves a pueblo San Ildefonso (_SAN), Pojoaque (_POJ), Nambe (_NAM), Tesuque(_TES) or Santa Fe County (_SFC) or if the purpose of the pipeline serves multiple entities then named (_RWA).
7. Metering
 - a. The stakeholders require special metering to allow the total water usage for each Pueblo to be determined, not included the non-tribal residents living within Pueblo boundaries. These non-tribal residents shall be individually metered, but the special metering system shall be designed to allow determination of Pueblo water usage without individually metering the Pueblo residents.
8. Ensure the system operation is not impeded by excluding, moving, or postponing installation of distribution lines.

1.08 ADDITIONAL BACKGROUND INFORMATION

A. Land Interests

1. Reclamation and Santa Fe County retain responsibility for obtaining land interests sufficient to construct the PBRWS. The contractor shall request necessary construction corridors from Reclamation, allowing six months for Reclamation to obtain private easements and access permission from the Pueblos. Land interests sufficient to construct will be obtained by Reclamation before Reclamation will issue a notice to proceed with construction.
 - a. Land interests include:
 - 1) Areas needed to install pipeline and pipeline appurtenances, bore pits, fire hydrants, air valves, air vents and meter pits, etc.
 - 2) Areas needed to restore, replace, and relocate existing features.
2. Contractor shall anticipate that there may be times when Pueblos temporarily restrict access to their lands with little notice.
3. Reclamation will transfer title of the PBRWS to the designated project stakeholders upon substantial completion of the entire PBRWS as directed by the Act, Section 611(h). In general, Reclamation will convey to: (1) each Pueblo, the portion of any Pueblo Water Facility that is located within the boundaries of the Pueblo; (2) the County, the County Distribution System; and (3) the Regional Water Authority, any remaining portions of the PBRWS. The entity receiving title to each project feature and pipeline segment is the “final owner”. The Act

and the Cost Sharing and System Integration Agreement describe how PBRWS costs will be allocated among the Federal government, the State of New Mexico and Santa Fe County. For the reasons stated in this paragraph, each project feature and each segment of transmission and distribution pipeline must have an associated and tracked cost and “final ownership” identifier.

B. Environmental Compliance

1. Reclamation will complete required environmental compliance activities before notice to proceed with construction is issued.
2. Is it assumed that notices to proceed will be issued sequentially, as sufficient land interests are acquired, and the environmental compliance activities are completed.

C. Cultural Resource Protection

1. Reclamation, Santa Fe County, and Tribal Historic Preservation Offices retain responsibility for monitoring and performing required cultural resource mitigation and treatment activities. Any unanticipated finds of cultural resources during construction, or unanticipated damages that occur to historic properties during construction will be handled in accordance with the Programmatic Agreement (PA) between Reclamation, the New Mexico State Historic Preservation Office, and project stakeholders as described in Section 01 57 90 of this document.
2. Contractor retains:
 - a. responsibility for incorporating the treatment plans identified in the PA in their work plans and
 - b. liability for damages to historic properties and cultural resources that occur during construction as a result of not following the treatment plans and protection measures identified in the PA. Contractors shall anticipate discoveries of cultural artifacts during construction that will cause work delays at that site and be prepared to work at alternative locations.

D. Final Owner

1. The Final Owner is the Pojoaque Basin Regional Water Authority (RWA). Santa Fe County is expected to act as Fiscal Agent for the RWA, upon completion of construction, acceptance and performance testing, commissioning, and the initial operations and maintenance period described in Section 01 86 46, which includes demonstration testing and operator training.

1.09 INFORMATIONAL REPORTS

- A. The Pojoaque Basin RWS feasibility design report provides descriptions, drawings, quantity estimates, and cost estimates for the proposed project at a feasibility level, as defined by Reclamation’s design and cost estimating activities Directives and Standards.
- B. Reclamation is providing the Feasibility Design Report as guidance only, data acquisition needs for 100% design are the responsibility of contractor. Some pipeline alignments,

storage tank, and pumping plant locations have not yet been finalized. Reclamation, the contractor, and the facilitator will work with the project stakeholders to finalize facility locations while minimizing adverse impacts to existing infrastructure and cultural resources.

1. Sites and alignments must be approved by the Pueblos.
 2. Additional data would need to be collected to develop 100% designs.
 3. Feasibility assumptions would need to be verified and confirmed
- C. In 2019, the project baseline was redefined to better fit within funding constraints. The resulting 611g Agreement and Consensus Design Concept provide additional guidance to understand stakeholders' preferences and design intent.
- D. In 2016 an Asset Inventory Report was completed which shows all utilities at that time to the Quality D Level for the entire project area.

1.10 SPECIFICATION REQUIREMENTS

- A. Imperative statements in these specifications are Contractor requirements, unless otherwise stated
- B. Where specifications are written in streamlined form, words "shall be" are included by inference where a colon (:) is used within sentence or phrase.

1.11 ACRONYMS

AAO	Albuquerque Area Office
AASHTO	American Association of State Highway and Transportation Officials
ACHP	Advisory Council on Historic Preservation
ACI	American Concrete Institute
ADA	Americans with Disabilities Act
ADCP	Acoustic Doppler Current Profiler
ANSI	American National Standards Institute
APS	Allowance for Procurement Strategies
ASR	Aquifer Storage and Recovery
ASTM	American Society for Testing and Materials
ATS	Automatic Transfer Switch
AWWA	American Water Works Association
BIA	Bureau of Indian Affairs
CAA	Clean Air Act
CE	Construction Engineer
CDMS	Construction Document Management System
CEQ	Council on Environmental Quality
CFR	Code of Federal Regulations
CIP	Clean in place
CLIN	Contract line-item number
CLSM	Controlled Low-Strength Material
CMMS	Computerized Maintenance Management System
CO	Contracting Officer
COR	Contracting Officer's Representative

Cost Sharing Agreement	Cost Sharing and System Integration Agreement
CT	Contact time
CWA	Clean Water Act
D/DBPR	Disinfectant/Disinfectant By-Product Rule
DB	design-build
DBB	design-bid-build
DBP	disinfection byproducts
DEC	Design Estimating Construction
DEIS	Draft environmental impact statement
DH	Drill Hole
DIAR	Department of Interior Acquisition Regulations
DNP	Distributed Network Protocol
DOC	Dissolved Organic Carbon
DOI	Department of the Interior
DTM	Digital Terrain Model
DWV	drain, waste, and vent system
EIS	Environmental Impact Statement
EPA	U. S. Environmental Protection Agency
EPDM	Ethylene Propylene Diene Monomer
ERI	electrical resistivity imaging
FAC-P/PM	Federal Acquisition Certification for Program and Project Managers
FAR	Federal Acquisitions Regulations
FBMS	Financial and Business Management System
FDR	Feasibility Design Report
FER	Field Exploration Requests

FOUO	For Official Use Only
FRP	fiberglass reinforced plastic.
FWCA	Fish and Wildlife Coordination Act
GAC	granular activated carbon
GIS	Geographic Information Systems
GMP	Guaranteed Maximum Price
GPS	Global Position System
HAA	haloacetic acids
HDD	Horizontal Directional Drilling
HDPE	high density polyethylene
HEC-RAS	Model name
HGL	Hydraulic grade line
HKM	Hurlbut, Kersich, and McCullough Engineers
HMI	Human Machine interface
HVAC	heating, ventilating, and air conditioning
IEC	International Electrotechnical Commission
IGCE	Independent Government Cost Estimates
ISEA	International Safety Equipment Association
JMEC	Jemez Mountain Electric Cooperative
LAN	Local Area Network
LiDAR	Light Detection and Ranging data
LT1ESTWR	Long Term 1 Enhanced Surface Water Treatment Rule
LT2ESTWR	Long Term 2 Enhanced Surface Water Treatment Rule
MCE	Maximum Considered Earthquake
MCL	Maximum Contaminant Levels

MERL	Materials Engineering and Research Laboratory
MF	microfiltration
NAGPRA	Native American Graves Protection and Repatriation Act
National	National Register of Historic Places Register
NDMA	N-Nitrosodimethylamine
NEC	National Electric Code
NEPA	National Environmental Policy Act
NFPA	National Fire Protection Association
NHPA	National Historic Preservation Act
NMAS	National Mapping Accuracy Standards
NMBGMR	New Mexico Bureau of Geology and Mineral Resources
NM DOT	New Mexico Department of Transportation
NMED	New Mexico Environment Department
NOM	natural organic material
NPDES	National Pollutant Discharge Elimination System
NPDWR	National Primary and Secondary Drinking Water Regulations
NTP	Notice to Proceed
NTU	Nephelometric Turbidity Units
ODBC	Open Database Connectivity
OMB	Office of Management and Budget
OM&R	Operation, Maintenance, and Replacement
O&M	Operations and Management
OSHA	Occupational Safety and Health Administration
PBRWS	Pojoaque Basin Regional Water System
PHA	Peak Horizontal Acceleration

PMT	Project Management Team
PNM	Public Service of New Mexico
PP	pumping plant
PVC	polyvinyl chloride
PVDF	polyvinylidene fluoride
PWL	Pumping Water Level
PWS	Performance Work Statement
RAR	Reclamation Acquisitions Regulations
Reclamation	Bureau of Reclamation
RFQ	Request for Quotation
RFP	Request for Proposal
ROD	Record of Decision
ROW	right(s) of way
ROW GOE	Right-of-way grants of easement
RSN	Required Submittal Number
RTK	Real Time Kinematic
RWA	Regional Water Authority
RWS	Regional Water System
SCADA	Supervisory Control and Data Acquisition
SDWA	Safe Drinking Water Act
Settlement Act	Title VI of the Claims Resolution Act of 2010
Settlement Act Pueblos	Pueblo de San Ildefonso and Pojoaque, Nambé, and Tesuque Pueblos
SHPO	State Historic Preservation Office
SOC	synthetic organic compounds
SOP	Standard Operating Procedures

STP	Standard Penetration Tests
SQL	Structured Query Language
SWL	Static Water Level
SWTR	Surface Water Treatment Rule
TCP/IP	Transmission Control Protocol/Internet Protocol
TCR	Total Coliform Rule
TDH	Total Dynamic head
THM	Trihalomethanes
TOC	Total Organic Carbon
TSC	Technical Services Center, Bureau of Reclamation, Denver Office
TSS	total suspended solids
UC	Upper Colorado Region
UF	ultrafiltration
UL USC	United States Code
USGS	United States Geologic Survey
USACE	United States Army Corps of Engineers
UV	ultraviolet
VAC	volts alternating current
VDC	volts direct current
VFD	Variable Frequency Drive
WBS	Work Breakdown Structure
WR&C	Water Removal & Control
Y	wye

1.12 MEASUREMENTS

°F	degrees Fahrenheit
----	--------------------

afm	acre-feet per month
afy	acre-feet per year
cfs	cubic feet per second
ea	each
f'c (psi)	compressive strength of concrete
fps	feet per second
ft	feet
ft ²	square feet
ft ³	cubic feet
fpd	feet per day
gal	gallon
gpd/lf	gallons per day per linear foot
gpd	gallons per day
gph	gallons per hour
gpm	gallons per minute
gpm/ft ²	gallons per minute per square foot
hp	horsepower
hrs	hours
in	inch
kPa	kilopascal
kVA	kilovolt amperes
kW	kilowatts
kWh	kilowatt hour
lb/ft ²	pounds per square foot
lb/ft ³	pounds per cubic foot

mg/kg	million gallons per kilogram
mgd	million gallons per day
mg/L	milligrams per liter
MHz	megahertz
mm	millimeter
msl	mean sea level
mNTU	milli Nephelometric Turbidity Unit
MWh	megawatts hour
ohm-cm	ohms per centimeter per cubic foot (pcf)
ppm	part per million
psi	pounds per square inch
rpm	revolutions per minute
sec-1	second-1
tsf	tons per square ft
yd2	square yards
yd3	cubic yards
W	watts
w/cm	maximum water-cementitious material ratio
ω 2	motor/pump momentum
μ M	micrometer
V	volts

1.13 SYMBOLS

°	degrees
G	velocity gradient
K	Hydraulic conductivity

Kp	Passive Earth Pressure Coefficient
Q	Flows
Ro	radius of influence
ωr^2	motor/pump momentum

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

This page intentionally left blank.